

OMIP-121: Immune Phenotyping of Canine Peripheral Leukocytes by Mass Cytometry

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ABSTRACT

Dogs are a critical non-rodent species used in preclinical safety studies, particularly, in the pharmaceutical field, due to their physiological, metabolic, and immunological similarities to humans. As such, immunophenotyping of canine peripheral blood mononuclear cells (PBMCs) plays a crucial role in translational research, immune monitoring, and safety evaluations in drug development. However, the limited availability of canine-specific antibodies restricts detailed and accurate immune profiling, which is essential for advancing safety evaluations in drug development. To address this challenge, we developed a 15-marker panel for comprehensive mass cytometry-based immunophenotyping of cryopreserved canine PBMCs. This panel encompasses major leukocyte subsets, including B cells, CD4+ T helper cells, regulatory T cells, CD8+ cytotoxic T cells, memory T cell subsets, natural killer T cells, natural killer cells, dendritic cells, CD4+ monocytes, classical monocytes, and neutrophils. We utilized both extracellular and intracellular markers to facilitate in-depth immune profiling, despite the limited availability of canine-specific antibodies. The panel was thoroughly optimized in terms of marker selection, antibody clone validation, and metal isotope pairing. Additionally, by the use of mass cytometry, several channels remain unoccupied, providing flexibility for future panel expansion.

1 | Panel Presentation

1.1 | Purpose and Appropriate Sample Types

The canine immune system exhibits several parallels with the human immune system, enhancing the translatability of immune profiling data from dogs to humans [1]. Insights gained from canine immunophenotyping would contribute to improving the effectiveness of comparative oncology by facilitating a deeper understanding of cancer biology and supporting the development of improved diagnostics and therapies applicable to both species. One of the key approaches for understanding immune responses in such contexts is the immunophenotyping of PBMCs. Mass cytometry enables high-dimensional analysis of immune cell populations, providing a comprehensive and precise characterization

at the single-cell level [2]. Previous advances—such as multicolor flow panels—have demonstrated the value of comprehensive immune profiling in dogs, while also highlighting the need for flexible, robust solutions for detailed leukocyte analysis [3, 4]. The implementation of mass cytometry allows the generation of a backbone panel, to which additional markers can be added without extensive panel design or modification of the backbone itself. Therefore, we developed a 15-marker mass cytometry panel designed for robust immunophenotyping of cryopreserved canine PBMCs (Tables 1 and 2). This panel enables the identification of multiple immune cell lineages, including B cells, CD4+ T cells, regulatory T cells (Tregs), CD8+ T cells, natural killer T (NKT) cells, natural killer (NK) cells, dendritic cells (DCs), CD4+ monocytes, classical monocytes, and neutrophils. Moreover, it provides an in-depth characterization of T cell differentiation states,

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distinguishing naïve, central memory, and effector memory phenotypes. CD45 was incorporated into the panel as a pan-leukocyte marker to confirm leukocyte identity and to support future applications involving canine nonlymphoid tissue analysis.

1.2 | Background

A comprehensive literature review was conducted to identify published studies on canine immunophenotyping [5]. In dogs, NKT cells were distinguished from conventional T cells based

on CD94 expression, as anti-canine CD94 monoclonal antibody has been demonstrated to effectively enrich NK/NKT cell populations from PBMCs [6]. Our analysis revealed that the major subset of NKT cells in healthy dogs is composed of CD8+ NKT cells (Figure 1). To characterize T cell subsets, CD4 and CD8 were employed to enable the identification of helper and cytotoxic T cells, respectively. Tregs were defined by co-expression of CD25 and FOXP3 [7]. Granzyme B was included to assess the cytotoxic activity of CD8+ T cells. To characterize the differentiation status of T cells, we employed Eomes/CD44 phenotypic paradigms to define naïve (CD44–Eomes–), central memory (CD44+ Eomes–), and effector memory (CD44+ Eomes+) T cell subsets [8–11]. Eomes, a T-box transcription factor best known for its role in cytotoxic lymphocyte differentiation [9, 12, 13], but which also exerts important functions in CD4+ T cell differentiation [8, 11]. Eomes is transcriptionally upregulated during T cell activation and effector differentiation, with highest expression in effector memory CD8+ T cells and long-lived memory precursors, distinguishing them from naïve or central memory cells that express lower levels [9, 13]. In parallel, CD44 is a well-established marker of antigen-experienced cells and has been used to identify memory T cells in dogs [10, 14]. CD44 is upregulated following antigenic/mitogenic activation and remains elevated on memory CD4+ and CD8+ T cells, which produce higher levels of effector cytokines such as TNF- α compared with CD44-low naïve cells [4, 14]. For the identification of B cells, we employed a combination of CD21, CD22, and MHC-II markers, which together provide a robust definition of the peripheral B cells [15]. Canine NK cells are phenotypically defined as CD3-CD21-GranzymeB+CD94+ [6, 16], and can be

TABLE 1 | Summary table for application of OMIP-121.

Purpose	Comprehensive characterization of the principal canine leukocyte subsets including B cells, CD4+ T helper cells, regulatory T cells, CD8+ cytotoxic T cells, memory T cell subsets, NKT cells, NK cells, dendritic cells, CD4+ monocytes, classical monocytes, and neutrophils by mass cytometry
Technology	Mass cytometry
Application	Research
Species	Dog
Cell type(s)	PBMCs
Cross-references	OMIP 065

TABLE 2 | Panel reagents used for OMIP-121.

Specificity	Clone	Target species	Metal isotope	Purpose
CD45	YKIX716.13	Dog	¹⁴¹ Pr	Pan leukocytes
CD4	YKIX302.9	Dog	¹⁵⁰ Nd	Helper T cells, monocyte subset, neutrophils
CD335	48A	Human	¹⁵¹ Eu	NK cells subset
CD21	CA2.1D6	Dog	¹⁵² Sm	B cells
CD25	P4A10	Dog	¹⁵³ Eu	Regulatory T cells, activated T cells
CD22	RFB-4	Human	¹⁵⁹ Tb	B cells
CD14	M5E2	Human	¹⁶⁰ Gd	Monocytes
Eomes	WD1928	Human	¹⁶³ Dy	Transcription factor for effector memory T cells and NK cells
CD44	YKIX337.8.7	Dog	¹⁶⁴ Dy	Central memory and effector memory T cells
FoxP3	FJK-16s	Mouse/rat	¹⁶⁵ Ho	Transcription factor for regulatory T cells
CD8a	YCATE55.9	Dog	¹⁶⁷ Er	Cytotoxic T cells, NKT cells, DCs, NK cells subset
Granzyme B	GB11	Human	¹⁷¹ Yb	Cytotoxic activity, NK cells
MHC-II	YKIX334.2	Dog	¹⁷³ Yb	Antigen-presenting cells
CD94	8H10	Dog	¹⁷⁵ Lu	NKT cells, NK cells
CD3	CA17.2A12	Dog	¹⁷⁶ Yb	Pan T cells
Cell-ID Intercalator	N/A	N/A	^{191/193} Ir	Single nucleated cell/doublets discrimination
Cell-ID Cisplatin	N/A	N/A	¹⁹⁵ Pt	Live/dead discrimination

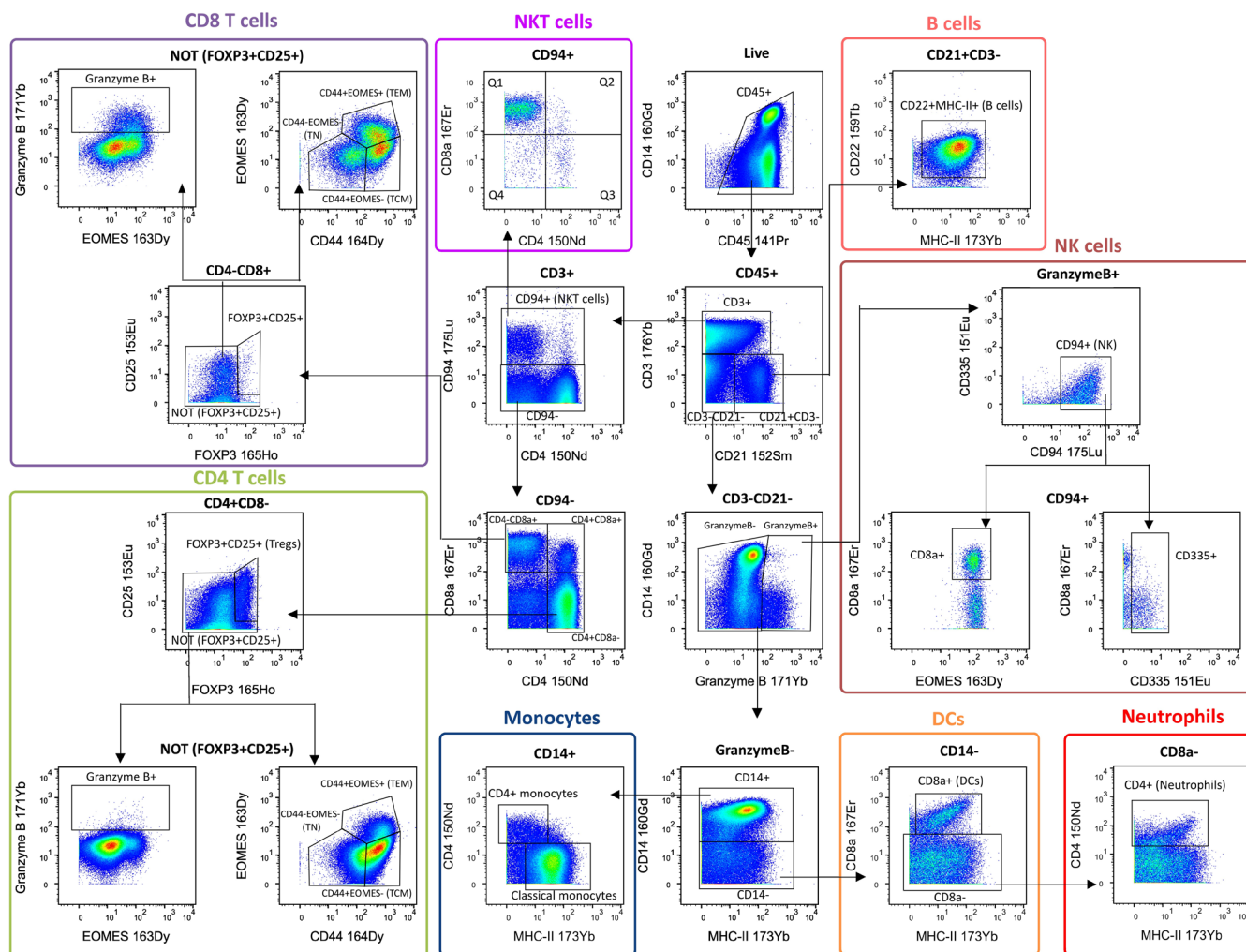


FIGURE 1 | Example staining and gating for OMIP-121. Representative example of manual gating strategy for cryopreserved canine PBMCs using CyTOF XT mass cytometer. Following data cleaning—including EQ calibration beads removal and pregating on live, singlet cells (Figure 9), leukocytes were first identified based on CD45 expression. T and B cells were subsequently identified using CD3 and CD21, respectively. B cell identity was further confirmed by CD22 and MHC-II expression (CD22+MHC-II+). Within the CD3+ population, NKT cells were defined as CD94+, whereas conventional T cells were defined as CD94-. Expression of CD4 and CD8 was assessed on NKT cells. Conventional T cells were further classified into single positive CD4+ T cells (CD4+CD8-), single positive CD8+ T cells (CD4-CD8+), and double positive (CD4+CD8+) subsets. In both CD4+CD8- and CD4-CD8+ T cells, regulatory T cells were identified as CD25+FOXP3+. Within the NOT (FOXP3+CD25+) population, CD44 and EOMES were used to classify naïve (CD44-EOMES-), central-memory (CD44+EOMES-), and effector-memory (CD44+EOMES+) subsets among CD4 and CD8 T cells. Expression of cytotoxic activity (Granzyme B) was also assessed on CD4 and CD8 conventional T cells. B cells were identified as CD21+CD22+MHC-II+. Within the CD3-CD21- population, NK cells were defined as Granzyme B+ and further characterized based on the expression of CD94. Expression of CD8a and CD335 were also assessed on NK cells. Within the Granzyme B- population, monocytes were identified as CD14+ and further subdivided into CD4+ monocytes (CD4+MHC-II-) and classical monocytes (CD4-MHC-II+). Within the CD14- population, DCs were defined as CD8+MHC-II+. Within the CD8- population, neutrophils were identified as CD4+. Mass cytometry data were exported as FCS file using ArcSinh transformation with a cofactor of 4. Analysis of mass cytometry data was carried out using FlowJo v10.10. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

further subdivided into CD335+ and CD335- subsets, which are proposed to represent different activation stages [17, 18]. The CD335+ subset has been shown to exhibit higher cytotoxic activity compared to the CD335- subset, although both populations display comparable cytokine secretion profiles [17, 18]. In addition, CD8 expression is frequently observed on canine NK cells, with reported frequencies reaching up to 30% under certain conditions [18–20]. Interestingly, we found that all CD8+ NK cells lacked CD335 expression, suggesting potential functional and developmental heterogeneity within the canine NK cell compartment (Figure 1). In addition to classical monocytes

(CD4–MHC-II+), dogs possess a distinct CD4+ monocyte subset that lacks MHC-II expression and constitutes a significantly smaller proportion of total CD14+ monocytes [21, 22]. Canine DCs exhibit morphological similarity to human DCs and are characterized by surface expression of CD8 and MHC-II [23]. Notably, CD4 expression was also detected on peripheral blood neutrophils, a unique feature among mammalian species [24]. All panel optimization and antibody titration experiments were performed on a CyTOF XT mass cytometer (Standard BioTools, South San Francisco, USA). The final panel was also acquired on a Lunarion mass cytometer (Polaris, Shanghai, China) to

assess panel performance and ensure compatibility across instruments.

Mass cytometry is a powerful tool for high-dimensional and high-throughput single-cell analysis, enabling the simultaneous measurement of up to 60 parameters within a single panel [2]. In addition to the availability of preconjugated antibodies, the pairing of antibodies with appropriate metal isotopes requires careful consideration to ensure optimal signal intensity while minimizing signal interference. Antibodies targeting low-abundance markers should be conjugated to isotopes within the instrument's high-sensitivity detection range, whereas highly abundant markers can be assigned to less sensitive channels [25]. Although signal spillover is substantially reduced in mass cytometry compared to fluorescent-based methods, three primary sources of interference must be addressed: $M \pm 1$ abundance spillover, $M + 16$ oxidation spillover, and isotope impurity [2]. To minimize these effects, low-abundance markers should not be assigned to the $M \pm 1$ or $M + 16$ channels of a highly expressed marker. Detailed steps of panel optimization are provided in Section 3.

1.3 | Similarity to Other Published OMIPS

To date, only one OMIP has focused on immunophenotyping of canine PBMCs and in-depth analysis of T-cell responses [3]. That panel comprised 13 markers and was designed for conventional flow cytometry. Our 15-marker panel represents the first mass cytometry-based OMIP designed for the comprehensive characterization of leukocyte subsets within canine PBMCs. While there is partial overlap with OMIP-065, which enables profiling of B cells, CD4 helper T cells, CD8 cytotoxic T cells, Tregs, NK cells, neutrophils, and monocyte subsets, our panel is optimized for the additional characterization of DCs, NKT cells, activated NK cells, and memory T cell subsets.

2 | Technical Information

For CyTOF XT, instrument tuning was performed using EQ Six Element Calibration Beads. This step calibrates the instrument for optimal sensitivity and mass accuracy. Calibration beads were automatically added to each sample before acquisition, ensuring consistent performance across runs. For Lunarion, instrument tuning was performed using LunaMix Tuning Solution. Instrument settings of both instruments were summarized in Table 3. Isotope-conjugated antibody used in the final OMIP are listed in Table 4.

2.1 | Materials

2.1.1 | Reagents

1. Cell-ID Cisplatin-195Pt, 100 μ L/CAT#201195|Standard BioTools
2. Fetal Bovine Serum, FBS|CAT#10500-064|Gibco

TABLE 3 | Instrument configuration for CyTOF XT and Lunarion mass cytometer.

Instrument	CyTOF XT
Channels	135
Mass range	75–209 amu
CyTOF software	Version 9.0.0 + 19045
Flow rate	30 μ L/min
Cell concentration	0.6×10^6 cells/mL
Sample carrier	Maxpar Cell Acquisition Solution Plus
Sample throughput	500 events per second
Calibration	EQ Six Element Calibration Bead
Instrument	Lunarion
Channels	140
Mass range	70–209 amu
Lunarion software	Web interface
Flow rate	80 μ L/min
Cell concentration	0.3×10^6 cells/mL
Sample carrier	LunaAcq Cell Acquisition Solution
Sample throughput	500–1000 events per seconds
Calibration	LunaMix tuning solution

3. Canine Fc Receptor Binding Inhibitor|CAT#14-9162-42|Fisher Scientific
4. Cell-ID Intercalator-Ir 125 μ M|CAT# 201192A|Standard BioTools
5. Maxpar Cell Acquisition Solution PLUS, CAS PLUS|CAT#201244|Standard BioTools
6. Maxpar Cell Staining Buffer, CSB|CAT# 201068|Standard BioTools
7. Maxpar Fix and Perm Buffer, 100 mL|CAT#201067|Standard BioTools
8. PBS 1 $\times$ without Ca^{2+} / Mg^{2+} |CAT# BE17-516F|Westburg
9. Pierce 16% Formaldehyde (w/v), Methanol-free|CAT#10751395|Fisher Scientific
10. eBioscience Foxp3/Transcription Factor Staining Buffer Set|CAT#00-5523-00|Fisher Scientific
11. DMSO|CAT#D2650-100ML|Sigma Aldrich
12. Ficoll Histopaque-1077 (1.077 g/mL)|CAT#17144001|GE Life Sciences
13. CASYton|CAT#5651808|OLS
14. LunaAcq Cell Acquisition Solution 10 $\times$ |PB011|Polaris Biology

TABLE 4 | Commercial panel reagents used for OMIP-121.

Specificity	Clone	Target species	Vendor	Catalog number	RRID	Uniprot entry name	Pre-conjugated/ in-house conjugated	Metal isotope	Dilution	Staining step
CD45	YKIX716.13	Dog	Bio-rad	MCA1042GA	AB_3099901	CD45: A0A8C0PL74, PTPRC_CANLF	In-house conjugated	¹⁴¹ Pr	1:300	Surface
CD4	YKIX302.9	Dog	Bio-rad	MCA1038GA	AB_3099890	CD4: P33705, CD4_CANLF	In-house conjugated	¹⁵⁰ Nd	1:300	Surface
CD335	48A	Human	Merck	MABF2109	N/A	CD335: O76036, NCR1_HUMAN	In-house conjugated	¹⁵¹ Eu	3:100	Surface
CD21	CA2.1D6	Dog	Bio-rad	MCA178IR	AB_323665	N/A	In-house conjugated	¹⁵² Sm	1:900	Surface
CD25	P4A10	Dog	Bio-rad	MCA5916GA	AB_3101154	CD25: Q28268, IL2RA_CANLF	In-house conjugated	¹⁵³ Eu	1:900	Surface
CD22	RFB-4	Human	Bioconnect	ANC-171-020	N/A	CD22: P20273, CD22_HUMAN	In-house conjugated	¹⁵⁹ Tb	1:100	Surface
CD14	M5E2	Human	Standard Biotools	3160001C	AB_2687634	CD14: P08571, CD14_HUMAN	Pre-conjugated	¹⁶⁰ Gd	1:100	Surface
Eomes	WDI928	Human	Invitrogen	15597446	AB_2572882	Eomes: O95936, EOMES_HUMAN	In-house conjugated	¹⁶³ Dy	3:100	Intracellular
CD44	YKIX337.8.7	Dog	Bio-rad	MCA1041GA	AB_3099900	CD44: Q28284, CD44_CANLF	In-house conjugated	¹⁶⁴ Dy	1:300	Surface
FoxP3	FJK-16s	Mouse/rat	Standard Biotools	3165024C	AB_2687843	Foxp3: Q99JB6, FOXP3_MOUSE	Pre-conjugated	¹⁶⁵ Ho	3:100	Intracellular
CD8a	YCATE55.9	Dog	Bio-rad	MCA1039GA	AB_3099899	CD8A: P33706, CD8A_CANLF	In-house conjugated	¹⁶⁷ Er	1:300	Surface
Granzyme B	GB11	Human	Standard Biotools	3171002B	AB_2687652	GZMB: P10144, GRAB_HUMAN	Pre-conjugated	¹⁷¹ Yb	3:100	Intracellular
MHC-II	YKIX334.2	Dog	Bio-rad	MCA1044GA	AB_11203404	MHC class II DR alpha chain: Q30437, Q30437_CANLF	In-house conjugated	¹⁷³ Yb	3:100	Surface

(Continues)

TABLE 4 | (Continued)

Specificity	Clone	Target species	Vendor	Catalog number	RRID	Uniprot entry name	Pre-conjugated/ in-house conjugated	Metal isotope	Dilution	Staining step
CD94	8H10	Dog	Bio-rad	MCA6400GA	AB_3101422	CD94: Q38HS3, KLRD1_CANLF	In-house conjugated	¹⁷⁵ Lu	3:100	Surface
CD3	CA17.2A12	Dog	Bio-rad	MCA1774GA	AB_1125250	CD3ε: P27597, CD3E_CANLF	In-house conjugated	¹⁷⁶ Yb	3:100	Surface

Abbreviation: N/A: not available.

2.1.2 | Buffers and Solutions

1. FACS Buffer 5%: 475 mL of PBX 1× + 25 mL of FBS
2. Freezing medium: 90% FBS + 10% DMSO
3. Cis-Pt Solution: Cis-Pt (5 mM) has to be diluted in PBS 1× to a final concentration of 1 μM.
4. Ir-Intercalator Solution: Ir-Intercalator 125 μM has to be diluted in MaxPar Fix and Perm Buffer to a final concentration of 50 nM.
5. 1.6% Formaldehyde Solution: Dilute 16% Formaldehyde in PBS 1× to a final 1.6% concentration.
6. Foxp3 Fixation/Permeabilization working solution: Dilute 1 part of Foxp3 Fixation/Permeabilization Concentrate with 3 parts of Foxp3 Fixation/Permeabilization Diluent.
7. Permeabilization Buffer working solution: Dilute 1 part of 10× Permeabilization Buffer with 9 parts of distilled water
8. LunaAcq Cell Acquisition Solution 1×: Dilute 1 part of LunaAcq Cell Acquisition Solution 10× with 9 parts of distilled water.

2.1.3 | Plastic Consumables

1. BD Falcon centrifuge tube, 15 mL|CAT#734-0451|VWR
2. Polypropylene tubes with cap, 5 mL|CAT#734-0447|VWR
3. Polystyrene tubes with cell strainer cap, 5 mL|CAT#352235|VWR
4. 96-well U-bottom plate| CAT#650160|Greiner

2.2 | Protocol

2.2.1 | PBMC Isolations and Cryopreservation

1. Split whole blood into 50 mL falcon tube. The maximum blood volume per falcon tube should be 20 mL.
2. Add PBS 1× up to 35 mL and gently mix
3. Add slowly 13 mL of the Ficoll Histopaque-1077 at the bottom of the tube.
4. Centrifuge at 400g for 30 min at room temperature. Acceleration and brake should be at 0.
5. Remove 10 mL of the PBS/Plasma upper layer
6. Carefully transfer the lymphocyte layer into a 50 mL tube.
7. Add FACS buffer up to 35 mL to tube with the lymphocytes.
8. Centrifuge at 300g for 10 min at room temperature. Acceleration and brake should be at 7.
9. Discard the supernatant.
10. Repeat the washing step one more time.
11. Resuspend the cells in 1 mL of FACS Buffer. Count the cells.

- Centrifuge at 300g for 10 min at room temperature. Acceleration and brake should be 7.
- Discard the supernatant.
- Resuspend the cell pellet in the cold freezing medium to obtain 10^7 in 1 mL per vial.
- Freeze the samples in Mr. Frosty at -80°C . After 24 h, transfer the vials into nitrogen tank for long-term storage.
- Centrifuge at 500g for 5 min at room temperature. Discard the supernatant.
- Add 5 mL of PBS 1 $\times$.
- Centrifuge at 500g for 5 min at room temperature. Discard the supernatant.
- Resuspend the cells in 1 mL of PBS 1 $\times$. Count the cells

2.2.3.2 | Cell Viability Staining

- Resuspend 2×10^6 cells in 200 μL of 1 μM Cis-Pt solution.
- Incubate for 5 min at room temperature.
- Quench staining with FACS Buffer using 5 $\times$ the volume of the Cis-Pt solution volume.
- Centrifuge at 500g for 5 min at room temperature. Discard the supernatant.
- Resuspend 2×10^6 cells in 65 μL of CSB. Transfer the cell suspension into a well of a 96-well plate.

2.2.2 | Antibody Conjugation

In-house labeling of antibodies was performed using MAXPAR X8 Antibody Labeling Kit according to the manufacturer's instruction. Conjugation of antibodies was achieved through the maleimide-cysteine reaction (Michael addition). One hundred microgram of Ab were used per reaction.

2.2.3 | Cell Staining Protocol

2.2.3.1 | PBMC Thawing

- Thaw the PBMC aliquot in a 37°C water bath until a small ice cube is still visible.
- Slowly transfer the PBMC in 9 mL of warm FBS.

2.2.3.3 | Cell Surface Staining

- Add 5 μL of Canine Fc Receptor Binding Inhibitor Solution and incubate for 20 min at room temperature.
- Prepare 30 μL of cell surface staining antibody cocktail in CSB.

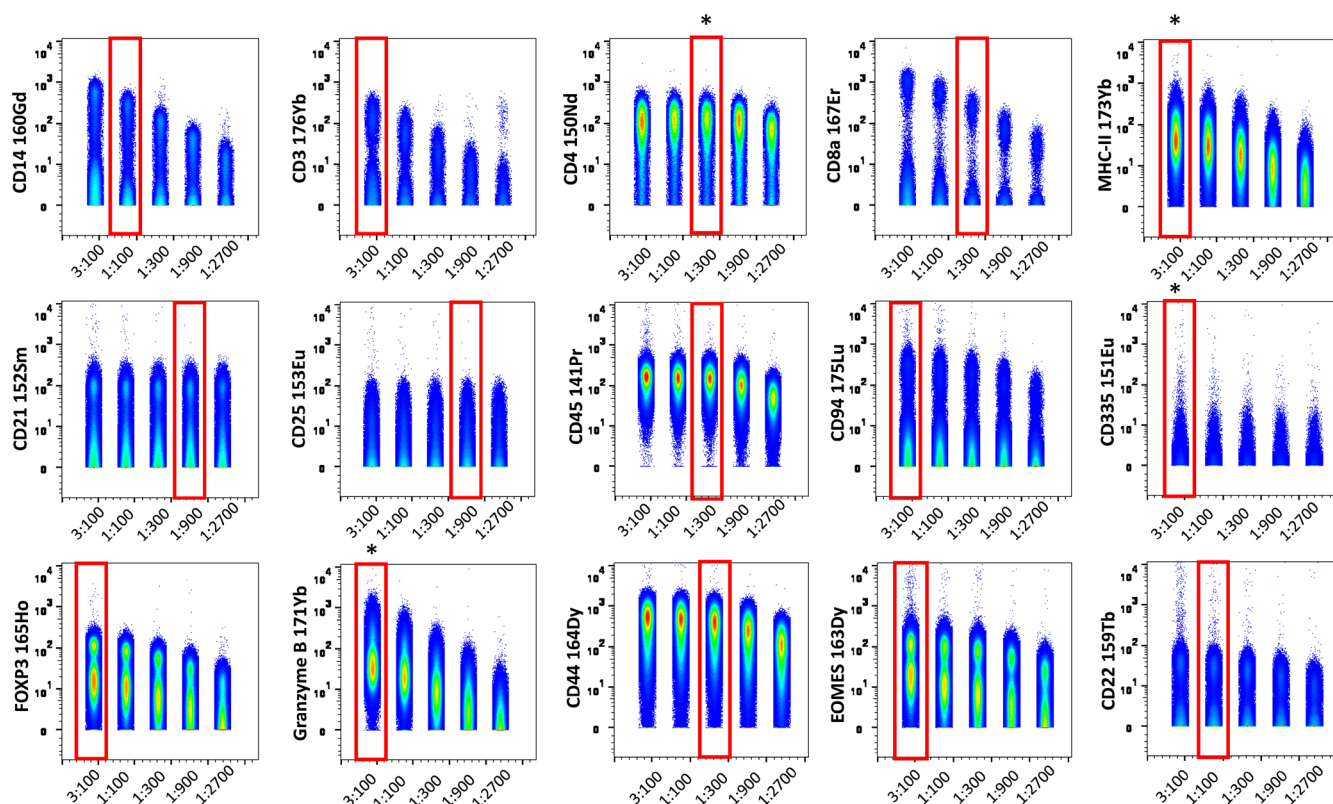


FIGURE 2 | Panel reagent titrations for OMIP-121. All antibodies were titrated on canine PBMCs using a threefold serial dilution series. Concatenated plots were gated on live single CD45+ cells. Optimal antibody concentrations are indicated by red boxes. Asterisks indicate where concatenated plots did not accurately reflect the optimal staining concentration; for these, individual bivariate plots gated on the most relevant cell population across the titration are provided in Figure 5. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

3. Add the antibody cocktail to the cell suspension for a final volume of 100 μ L. Mix sample gently.
4. Incubate for 30 min at room temperature.
5. Wash by adding 100 μ L of CSB to each well, mix gently, and centrifuge at 500g for 5 min at room temperature. Flick off the supernatant.
6. Repeat the washing step twice.

2.2.3.4 | Intracellular Staining

1. Add 200 μ L of FoxP3 Fixation/Permeabilization working solution to each well. Mix sample gently. Incubate for 45 min at 4°C.
2. Centrifuge at 800g for 10 min at 4°C. Flick off the supernatant.
3. Wash by adding 200 μ L of 1 $\times$ Permeabilization Buffer to each well.

4. Centrifuge at 800g for 10 min at 4°C. Flick off the supernatant.
5. Repeat the washing step once.
6. Resuspend cell pellet in 70 μ L of 1 $\times$ Permeabilization Buffer.
7. Prepare 30 μ L of intracellular staining antibody cocktail in Permeabilization Buffer 1 $\times$.
8. Add the antibody cocktail to the cell suspension for a final volume of 100 μ L. Mix sample gently.
9. Incubate for 1 h at 4°C.
10. Wash by adding 100 μ L of CSB to each well, mix gently, and centrifuge at 800g for 5 min at 4°C. Flick off the supernatant.
11. Repeat the washing step twice.
12. Add 200 μ L of 1.6% formaldehyde solution to each well. Mix sample gently.

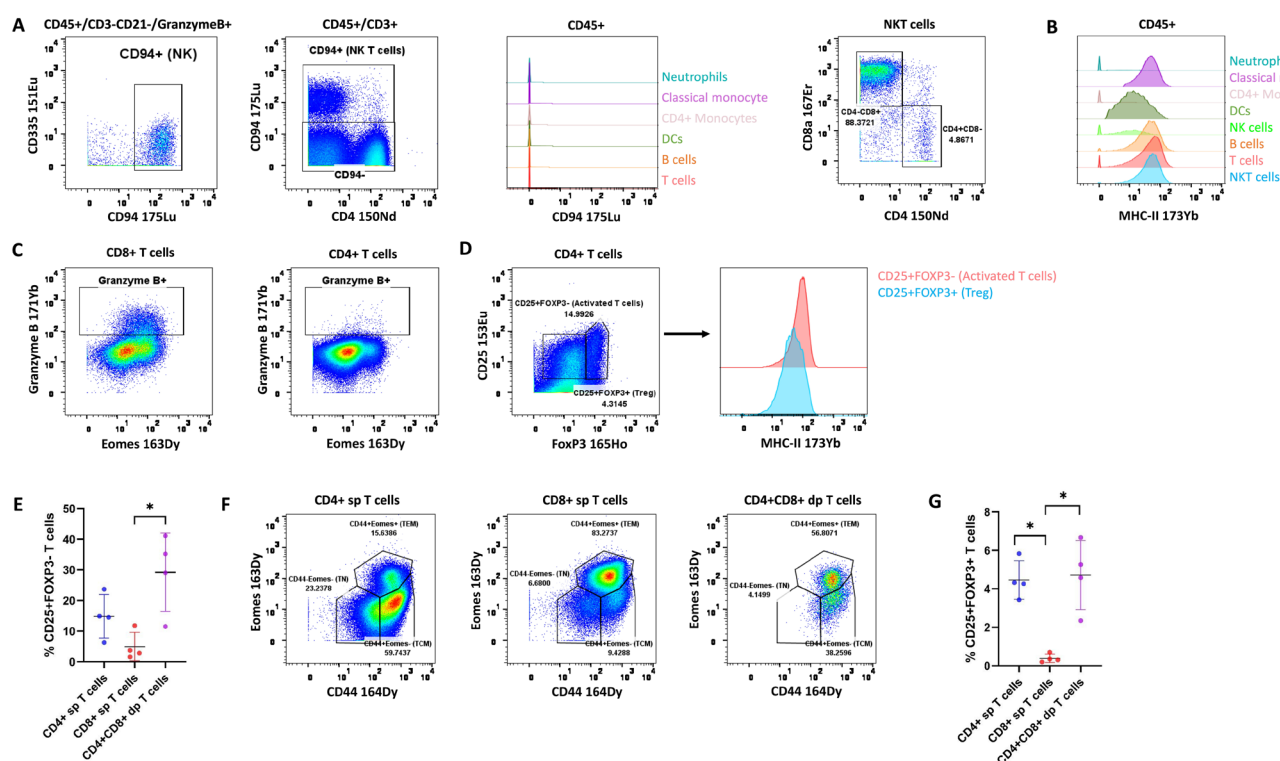


FIGURE 3 | Characterization of canine NKT and T cell subsets. (A) The first graph presents a representative plot of CD94 versus CD335 on CD45+/CD3-CD21-/GranzymeB+ to identify NK cells (CD94+). The second graph presents a representative plot of CD4 versus CD94 on CD45+/CD3+ to identify NKT cells (CD94+). The third graph presents a histogram depicting expression levels of CD94 on distinct leukocyte subsets defined in Table 5. The fourth graph presents a representative plot of CD4 versus CD8a expression on NKT cells, with frequencies of each subset indicated. (B) Histogram depicting expression levels of MHC-II on distinct leukocyte subsets defined in Table 5. (C) Representative plot of Granzyme B versus CD3 expression among CD8+ T cells and CD4+ T cells. (D) Representative plot of CD25 versus FOXP3 on CD4+ T cells to identify Tregs (CD25+FOXP3+) and activated T cells (CD25+FOXP3-). Corresponding histogram displays the expression level of MHC-II within these subsets. (E) Frequencies of activated T cells (CD25+FOXP3-) within CD4+ sp, CD8+ sp, and CD4+CD8+ dp T cells. Only significant comparisons are indicated: p (CD8+ sp. T cells vs. CD4+CD8+ dp T cells)=0.0337. (F) Representative plot of CD44 versus Eomes distinguishing naive (TN, CD44-Eomes-), central memory (TCM, CD44+Eomes-), effector memory (TEM, CD44+Eomes+) T cell population on CD4+ sp, CD8+ sp, and CD4+CD8+ dp T cells. (G) Frequencies of Tregs (CD25+FOXP3+) in CD4+ sp, CD8+ sp, and CD4+CD8+ dp T cells. Only significant comparisons are indicated: p (CD4+ sp. T cells vs. CD8+ sp. T cells)=0.0106, p (CD8+ sp. T cells vs. CD4+CD8+ dp T cells)=0.0256. For (E) and (G), each circle represents an individual dog. The results are shown as the mean $\pm$ SD ($n=4$ including D1, D2, D3, D4). Statistical analysis was performed with a repeated measures one-way ANOVA, followed by Tukey's post hoc tests. Statistically significant differences are indicated by asterisks (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

13. Incubate for 15 min at room temperature.
14. Centrifuge at 800g for 10 min at room temperature. Flick off the supernatant.
15. Add 200 μ L of the 50 nM Ir-Intercalator Solution to each well. Mix sample gently. Transfer all the cell suspension to a 5 mL polypropylene tube.

Add 800 μ L of the 50 nM Ir-Intercalator Solution to each tube to reach a final volume of 1 mL.

16. Incubate overnight at 4°C.

Note: If necessary, sample can be stored in the Ir-Intercalator Solution at 4°C for up to 5 days.

2.2.3.5 | Sample Acquisition

1. Wash by adding 2 mL of CSB. Centrifuge at 800g for 5 min at 4°C. Discard the supernatant.
2. Repeat this washing step once.
3. Wash by adding:
 - a. 2 mL of CAS PLUS for CyTOF XT acquisition.
 - b. 2 mL of LunaAcq Cell Acquisition Solution 1 $\times$ for Lunarion acquisition.
4. Centrifuge at 800g for 5 min at 4°C. Discard the supernatant.
5. Repeat this washing step once.
6. Resuspend the cell in the residual volume. Add:

- a. CAS PLUS to reach a final volume of 1 mL for CyTOF XT acquisition.
 - b. LunaAcq Cell Acquisition Solution 1 $\times$ to reach a final volume of 1 mL for Lunarion acquisition.
7. Filter the cells through a 35 μ m cell strainer
 8. Count the cells. Adjust the volume with:
 - a. CAS PLUS to reach a final concentration of 0.6×10^6 cells/mL for CyTOF XT acquisition. Add 90% of the calculated volume of CAS PLUS. The remaining 10% are the EQ Six Element Calibration Beads that will be added by the CyTOF XT.
 - b. LunaAcq Cell Acquisition Solution 1 $\times$ to reach a final concentration of 0.3×10^6 cells/mL for Lunarion acquisition.

3 | Panel Design and Optimization

This OMIP was developed to enable comprehensive characterization of the principal leukocyte subsets within canine PBMCs. This is the first mass cytometry OMIP specifically designed for the in-depth analysis of cryopreserved canine PBMCs. The panel was optimized through the selection of dog-specific and cross-reactive antibodies targeting B cells, CD4+ helper T cells, Tregs, CD8+ cytotoxic T cells, memory T cell subsets, NKT cells, NK cells, DCs, CD4+ monocytes, classical monocytes, and neutrophils. Additionally, marker and metal isotope pairing were selected based on the anticipated expression level of each

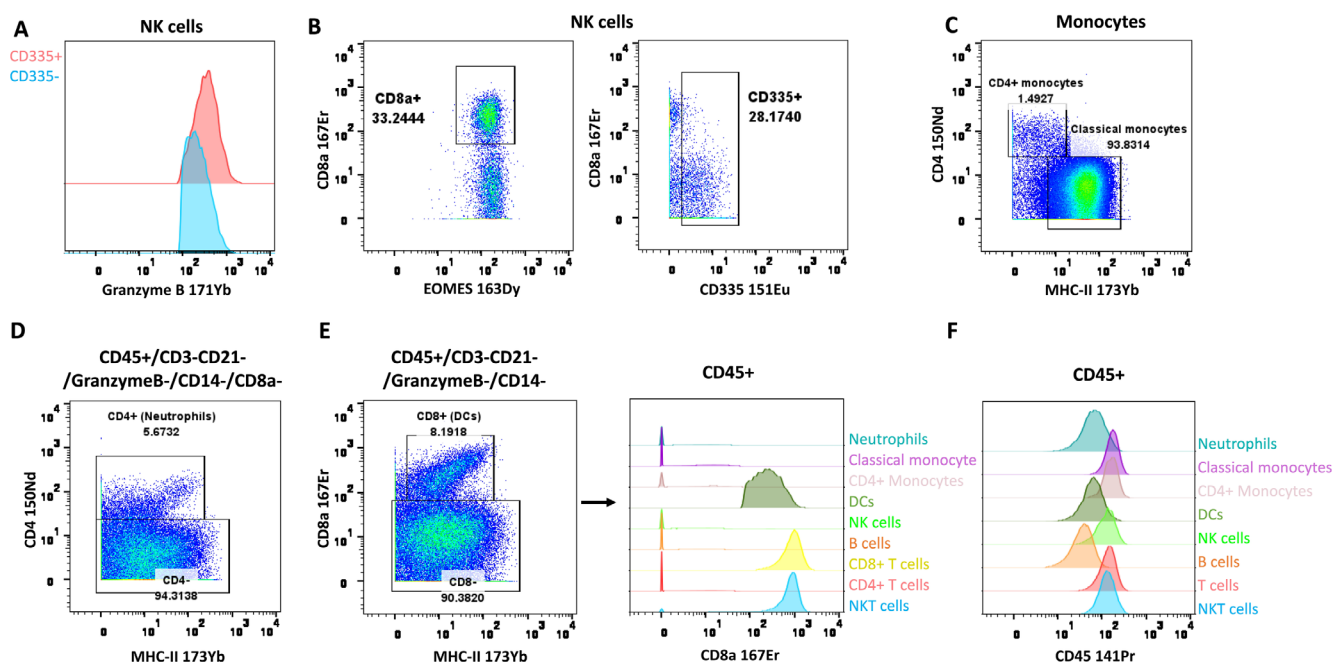


FIGURE 4 | Characterization of canine NK cells, monocytes, neutrophils, and DCs. (A) Histogram depicting expression levels of Granzyme B on CD335+ and CD335- NK cell populations. (B) Representative plot of Eomes versus CD8a and CD335 versus CD8a on NK cells. (C) Representative plot of MHC-II versus CD4 on monocytes to identify CD4+ monocytes (CD4+MHC-II-) and classical monocytes (CD4-MHC-II+). (D) Representative plot of MHC-II versus CD4 on CD45+/CD3-CD21-/GranzymeB-/CD14-/CD8a- population to identify neutrophils (CD4+). (E) Representative plot of MHC-II versus CD8a on CD45+/CD3-CD21-/GranzymeB-/CD14- population to identify DCs (CD8a+MHCII+). Corresponding histogram displays expression levels of CD8a on distinct leukocyte subsets defined in Table 5. (F) Histogram depicting expression levels of CD45 on distinct leukocyte subsets defined in Table 5. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

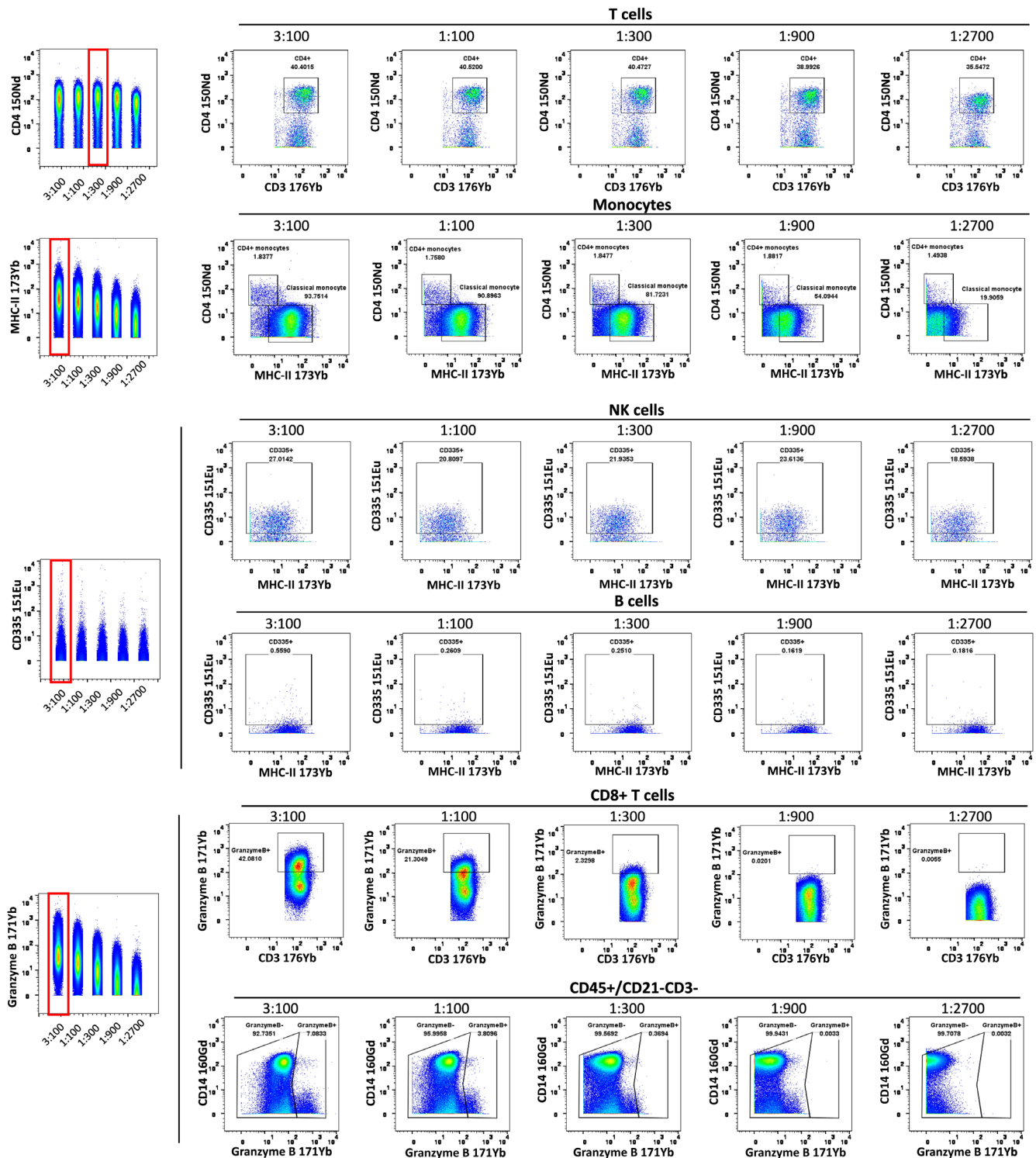


FIGURE 5 | Detailed analysis of markers for which concatenated plots did not accurately reflect the optimal staining concentrations. The first row shows CD4 antibody titration alongside a bivariate plot of CD3 versus CD4 within the T cell population across the serial dilution series. The second row shows MHC-II antibody titration alongside a bivariate plot of MHC-II versus CD4 within the monocyte population across the serial dilution series. The third row shows CD335 antibody titration alongside a bivariate plot of MHC-II versus CD335 within NK cells and B cells (negative control). The fourth row shows Granzyme B antibody titration alongside a bivariate plot of CD3 versus Granzyme B within CD8+ T cells and Granzyme B versus CD14 within the CD45+/CD21-CD3- population to identify NK cells. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

marker and the detection sensitivity of the corresponding metal isotope. Additional considerations included the availability and performance of both commercially available and in-house conjugated antibodies.

Panel construction commenced with a seven-parameter lineage marker “core” panel, as detailed in the Section 3.2. Building upon this “core” panel, additional antibodies were incorporated sequentially according to the general principles

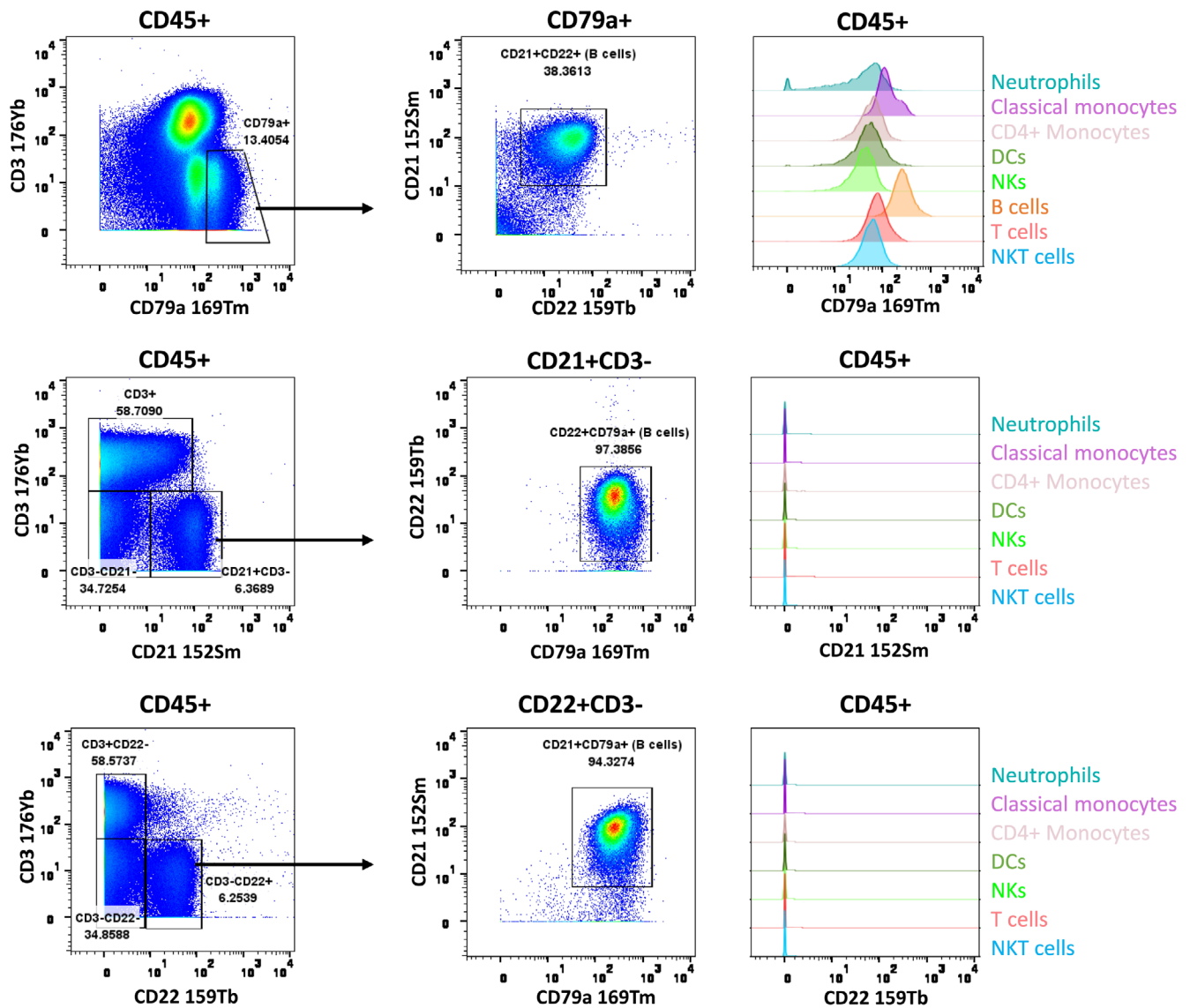


FIGURE 6 | Comparative analysis of CD21, CD22, and CD79a as B cell markers in canine PBMCs. The first row presents a representative plot of CD79a versus CD3 within the live single CD45+ population, illustrating gating of the CD79a+ population. Subsequent plots show the expression of CD22 and CD21 within this gated CD79a+ population to identify B cells. Accompanying histogram depicts CD79a expression levels across distinct leukocyte subsets defined in Table 5. The second row presents a representative plot of CD21 versus CD3 within the live single CD45+ population, illustrating gating of the CD21+CD3- population. Subsequent plots show the expression of CD79a and CD22 within this gated CD21+CD3- population to identify B cells. Accompanying histogram depicts CD21 expression levels across distinct leukocyte subsets defined in Table 5. The third row presents a representative plot of CD22 versus CD3 within the live single CD45+ population, illustrating gating of the CD22+CD3- population. Subsequent plots show the expression of CD79a and CD21 within this gated CD22+CD3- population to identify B cells. Accompanying histogram depicts CD22 expression levels across distinct leukocyte subsets defined in Table 5. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

of panel design in mass cytometry [2]. Commercially available isotope-conjugated antibodies were obtained from Standard BioTools (South San Francisco, CA, USA), while all other antibodies were conjugated in-house using the MAXPAR X8 Antibody Labeling Kit. Subsequently, each antibody was titrated, and optimal concentrations were determined for inclusion in the final panel (Figure 2). All panel optimization and antibody titration experiments were performed on the CyTOF XT. Upon finalization of the panel, stained samples were acquired on both the CyTOF XT and Lunatron instruments to assess panel performance and ensure compatibility across instruments.

3.1 | Panel Design

A comprehensive literature review was conducted to identify published studies on canine immunophenotyping and vendor websites for the most current dog-specific and cross-reactive antibodies available commercially [26]. CD45 was included in the panel as a pan-leukocyte marker to confirm leukocyte identity and to facilitate future use for the analysis of canine nonlymphoid tissue samples.

For initial gating, CD3 and CD21 were employed to discriminate T cells and B cells, respectively. B cell identity was further

confirmed by CD22 and MHC-II expression (CD22 + MHC-II+). Within the T cell population, CD94 was subsequently used to distinguish NKT cells from conventional T cells, as anti-CD94 monoclonal Ab has been demonstrated to effectively enrich NK/NKT cell populations from PBMCs [6]. This finding is consistent with our analysis, in which CD94 expression was restricted to the NK cell and NKT cell subsets and absent from all other leukocyte populations (Figure 3A). Our analysis revealed that the major subset of NKT cells is composed of CD8+ NKT cells (Figure 3A). T cell subsets were further defined based on CD4 and CD8 expression, enabling the identification of helper and cytotoxic T cell populations. Notably, canine T cells also express MHC-II (Figure 3B). Granzyme B was included to assess the cytotoxic activity of CD8+ T cells (Figure 3C), while Tregs were defined by co-expression of CD25 and FOXP3 [7] (Figure 3D). CD25 is also considered one of the functionally most important T cell activation markers [27]. We found that activated T cells (CD25 + FOXP3-) display higher MHC-II expression compared to Tregs (CD25 + FOXP3+) (Figure 3D), which is consistent with previous findings by Doveren et al. [28]. Additionally, a minor population of CD4 + CD8+ double-positive (dp) T cells, comprising approximately 2% of total T cells, displayed characteristics of activation including upregulation of CD25 relative to CD4+ and CD8+ single-positive (sp) T cells (Figure 3E) [10, 29]. Compared to CD4+ and CD8+ sp. T cells, dp T cells showed reduced frequencies of naïve cells and increased frequencies of central-memory and effector-memory T cells (Figure 3F). Notably, dp T cells also contain a subset of CD25 + FOXP3+ regulatory T cells, suggesting considerable heterogeneity within this population (Figure 3G) [10].

Canine NK cells are phenotypically defined as CD3-CD21-GranzymeB+CD94+, and can be further subdivided into CD335+ and CD335- subsets, which are proposed to represent different activation stages [6, 17, 18, 20, 30]. CD335 (also known as Nkp46 or NCR1) functions as an activating receptor on canine NK cells, initiating cytotoxic responses upon ligand engagement. CD335+ NK cells show spontaneous cytotoxicity against tumor targets like osteosarcoma cells in calcein release assays, which supports its use as a marker of functional maturity [18]. Moreover, the CD335+ subset expressed higher baseline levels of NK-associated genes (e.g., granzyme B, perforin) and transcription factors like GATA3, reflecting an activated state, while CD335- cells represent immature precursors that upregulate CD335 upon IL-2 stimulation or expansion [20]. Both subsets display comparable cytokine secretion profiles poststimulation [18]. In the present study, the CD335+ subset has been shown to exhibit higher cytotoxic activity compared to the CD335- subset (Figure 4A). Additionally, CD8 expression is frequently observed on putative canine NK cells, with the expression in some conditions reportedly encompassing as much as 30% of the population (Figure 4B) [18, 19]. Interestingly, all CD8+ NK cells identified lacked CD335 expression, suggesting potential functional or developmental distinctions within the NK cell compartment (Figure 4B).

In addition to classical monocytes (CD14+CD4-MHC-II+), dogs possess a CD4+ monocyte subset that lacks MHC-II expression and constitutes a significantly smaller proportion of total CD14+ monocytes [21, 22, 31] (Figure 4C). Interestingly, dogs are unique among mammals in exhibiting CD4 expression on peripheral blood neutrophils [24] (Figure 4D).

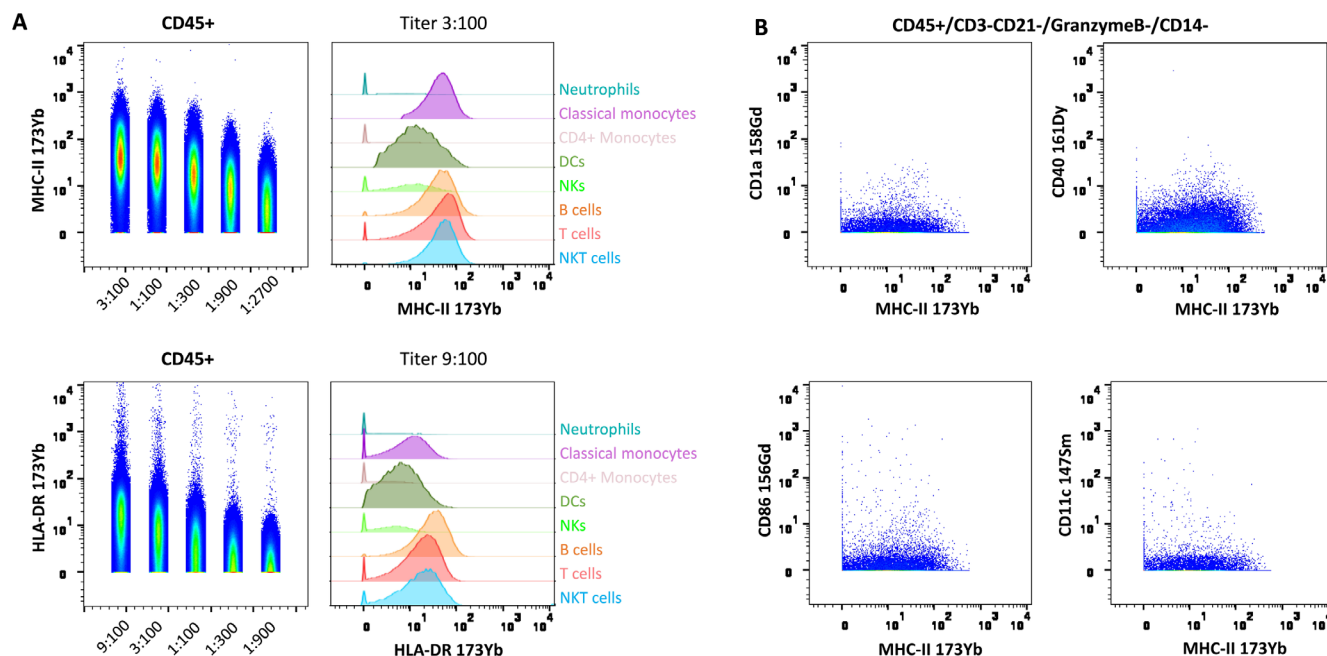


FIGURE 7 | (A) Comparative analysis of HLA-DR and MHC-II as antigen-presenting cells markers in canine PBMCs. (B) Characterization of canine DCs. (A) The first row shows the titration of MHC-II on canine PBMCs using a threefold serial dilution series. Concatenated plot was gated on live single CD45+ cells. Accompanying histogram depicts MHC-II expression levels across distinct leukocyte subsets defined in Table 5 at the antibody titer of 3:100. The second row presents the titration of HLA-DR on canine PBMCs using a threefold serial dilution series. Concatenated plot was gated on live single CD45+ cells. Accompanying histogram depicts HLA-DR expression levels across distinct leukocyte subsets defined in Table 5 at the antibody titer of 9:100. (B) Representative plot of MHC-II versus CD1a, CD40, CD86, CD11c on CD45+/CD3-CD21-/GranzymeB-/CD14- population. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Canine DCs exhibit morphological similarity to human DCs and are characterized by surface expression of CD8 and MHC-II [23]. Our data indicate that CD8 expression level on DCs is lower than that observed on CD8+ T cells and NKT cells (Figure 4E).

Finally, we investigated CD45 expression patterns across identified leukocyte subsets. B cells exhibited the lowest CD45 expression, followed by DCs and neutrophils, while classical monocytes, CD4+ monocytes, NK cells, NKT cells, and T cells all expressed comparably high levels of CD45 (Figure 4F).

3.2 | Panel Optimization

3.2.1 | Reagent Titration

Each commercial or in-house isotope-conjugated antibody was titrated on canine PBMCs to determine the optimal antibody concentration for labeling 2×10^6 cells in 100 μ L of cell staining

buffer (CSB) for extracellular antibody or Permeabilization Buffer for intracellular antibody. Antibody cocktails were freshly prepared before each staining procedure. For both commercial and in-house conjugates, serial titrations were performed using threefold dilutions at 3:100, 1:100, 1:300, 1:900, 1:2700. Optimal concentrations were selected based on the ability to resolve known cell subsets and to achieve clear separation between negative and positive populations, while also considering background signal and the frequency of the target population. Titration data of all antibodies included in the final panel are presented in Figure 2. In cases where concatenated plots did not accurately reflect the optimal staining concentration, individual bivariate plots gated on the most relevant cell population across the titration are shown in Figure 5.

3.2.2 | Testing of Alternative Reagents

Optimal pairing of antibodies with metal isotopes in mass cytometry requires careful consideration of both the sensitivity

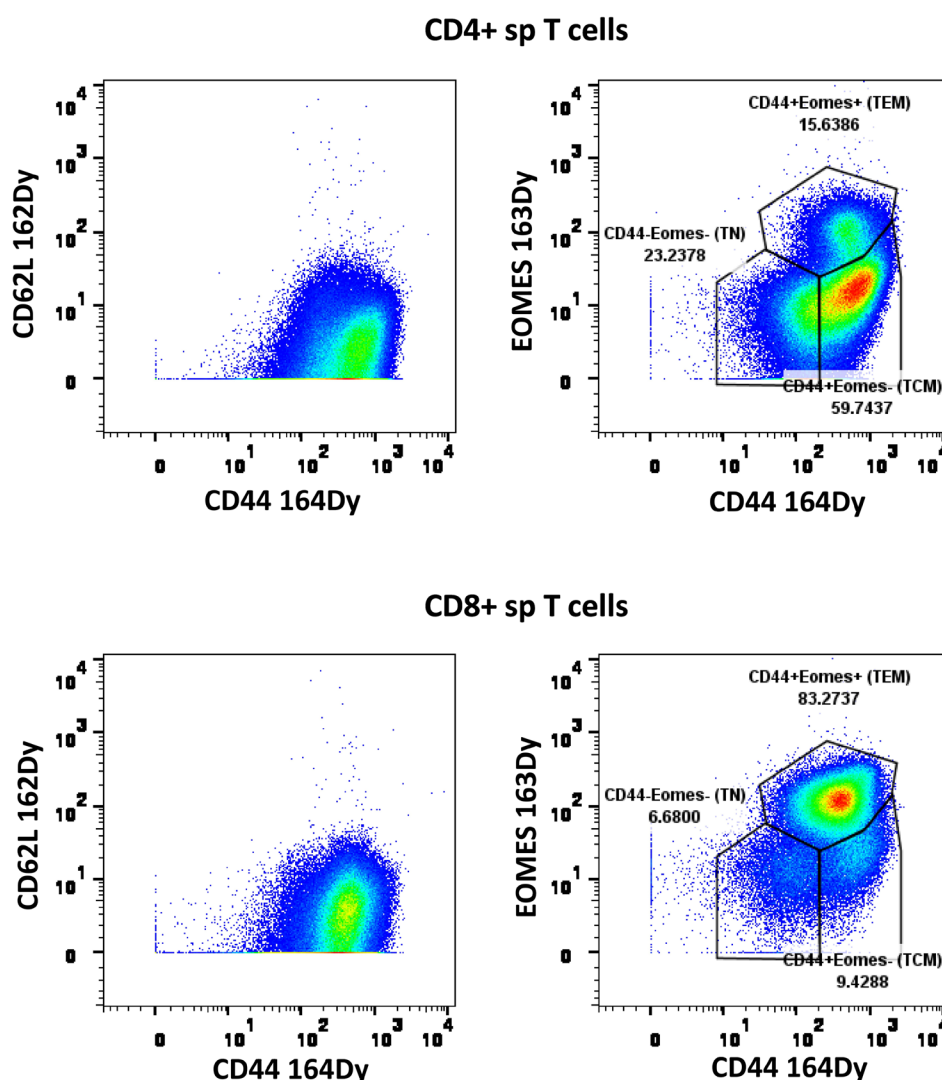


FIGURE 8 | Comparative analysis of CD44/CD62L and CD44/Eomes phenotypic paradigms for defining memory T cell subsets in canine CD4+ and CD8+ T cells. The first row shows the memory subset of CD4+ T cells, while the second row presents corresponding analysis for CD8+ T cells. The CD44 versus Eomes plots distinguish naïve (TN, CD44–Eomes–), central memory (TCM, CD44+Eomes–), and effector memory (TEM, CD44+Eomes+) T cell populations. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 5 | Phenotypic definition of cellular populations.

Phenotypic definition	Cell population
CD45+	Total leukocytes
CD45+/CD21+CD3−/ CD22+MHC-II+	B cells
CD45+/CD3+/CD94+	NKT cells
CD45+/CD3+/CD94−	T cells
CD45+/CD3+/CD94−/ CD4+CD8a−	CD4+ T cells (Helper T cells)
CD45+/CD3+/ CD94−/CD4+CD8a−/ CD25+FOXP3+	Regulatory T cells
CD45+/CD3+/ CD94−/CD4+CD8a−/ CD25+FOXP3−	Activated CD4+ T cells
CD45+/CD3+/ CD94−/CD4+CD8a−/ NOT(CD25+FOXP3+)/ CD44+Eomes+	Effector memory CD4+ T cells
CD45+/CD3+/ CD94−/CD4+CD8a−/ NOT(CD25+FOXP3+)/ CD44+Eomes−	Central memory CD4+ T cells
CD45+/CD3+/ CD94−/CD4+CD8a−/ NOT(CD25+FOXP3+)/ CD44−Eomes−	Naïve CD4+ T cells
CD45+/CD3+/CD94−/ CD4−CD8a+	CD8+ T cells (Cytotoxic T cells)
CD45+/CD3+/ CD94−/CD4−CD8a+/ CD25+FOXP3−	Activated CD8+ T cells
CD45+/CD3+/ CD94−/CD4−CD8a+/ NOT(CD25+FOXP3+)/ CD44+Eomes+	Effector memory CD8+ T cells
CD45+/CD3+/ CD94−/CD4−CD8a+/ NOT(CD25+FOXP3+)/ CD44+Eomes−	Central memory CD8+ T cells
CD45+/CD3+/ CD94−/CD4−CD8a+/ NOT(CD25+FOXP3+)/ CD44−Eomes−	Naïve CD8+ T cells
CD45+/CD3+/CD94−/ CD4+CD8a+	Double positive CD4 CD8 T cells
CD45+/CD3+/CD94−/ CD4−CD8a−	Double negative CD4 CD8 T cells

(Continues)

TABLE 5 | (Continued)

Phenotypic definition	Cell population
CD45+/CD21−CD3−/ GranzymeB+/CD94+	NK cells
CD45+/CD21−CD3−/ GranzymeB+/CD94+/ CD335+	Activated NK cells
CD45+/CD21−CD3−/ GranzymeB−/CD14+	Monocytes
CD45+/CD21−CD3−/ GranzymeB−/CD14+/ CD4+MHC-II−	CD4+ monocytes
CD45+/CD21−CD3−/ GranzymeB−/CD14+/ CD4−MHC-II+	Classical monocytes
CD45+/CD21−CD3−/ GranzymeB−/CD14−/ CD8a+	Dendritic cells
CD45+/CD21−CD3−/ GranzymeB−/CD14−/ CD8a−/CD4+	Neutrophils

range of the detection instrument and the expression level of the target antigen to maximize signal intensity while minimizing signal overlap. The CyTOF XT instrument demonstrates peak sensitivity for isotopes with atomic masses between 159 and 169; accordingly, isotopes within this range are preferentially assigned to Ab targeting low-abundance markers to enhance detection sensitivity. Although signal spillover is substantially reduced in mass cytometry compared to fluorescence-based methods, three primary sources of interference must be addressed: $M \pm 1$ abundance spillover, $M + 16$ oxidation spillover, and isotope impurity [2]. To minimize these effects, low-abundance targets should be avoided in the $M \pm 1$ or $M + 16$ channels of isotopes assigned to highly expressed markers.

Initially, a “core-panel” was established that enables the detection of major leukocyte subsets and facilitates feasibility studies and protocol optimization. This panel included anti-canine CD45 (Clone YKIX716.13, ^{141}Pr), anti-canine CD3 (Clone CA17.2A12, ^{176}Yb), anti-human CD14 (Clone M5E2, ^{160}Gd) [32], anti-canine CD4 (Clone YKIX302.9, ^{150}Nd), anti-canine CD8a (Clone YCATE55.9, ^{167}Er), anti-canine CD21 (Clone CA2.1D6, ^{152}Sm), and anti-canine MHC-II (Clone YKIX334.2, ^{173}Yb).

For the detection of B cells, we evaluated anti-human CD79a (Clone HM57, ^{169}Tm) as CD79a is reportedly expressed throughout all stages of B-cell development, while CD21 (Clone CA2.1D6, ^{152}Sm) and CD22 (Clone RFB-4, ^{159}Tb) are restricted to mature B cells [15]. However, our results showed that anti-human CD79a antibody did not exclusively stain B cells (Figure 6). Therefore, we have chosen anti-canine CD21 and anti-human CD22 to be the B-cell-specific antibodies, either of which can be combined

TABLE 6 | Panel iterations tested.

Specificity	Clone	Pre-conjugated/ in-house conjugated	Isotope	Staining	Target	Source	Catalog number	Reason for exclusion
CD4	SK3	Pre-conjugated	¹⁴⁴ Nd	Surface	Human	Standard Biotools	3144029C	No staining detected
CD62L	FMC46	In-house conjugated	¹⁴⁵ Nd	Surface	Human	Bio-rad	MCA1076GA	Poor resolution
CD11c	Bu15	Pre-conjugated	¹⁴⁷ Sm	Surface	Human	Standard Biotools	3147008C	No staining detected
CD16	3G8	Pre-conjugated	¹⁴⁸ Nd	Surface	Human	Standard Biotools	3148004C	No staining detected
CD123	107D2.08	In-house conjugated	¹⁵⁴ Sm	Surface	Human	Biotechnie	DDX0300P-100	No staining detected
CD86	Bu63	In-house conjugated	¹⁵⁶ Gd	Surface	Human	Bio-rad	MCA1118	Only under stimulation
CD1a	NA1/34	In-house conjugated	¹⁵⁸ Gd	Surface	Human	Bio-rad	MCA80GA	Only under stimulation
CD40	LOB7/6	In-house conjugated	¹⁶¹ Dy	Surface	Human	Bio-rad	MCA1590	Only under stimulation
CD62L	FMC46	In-house conjugated	¹⁶² Dy	Surface	Human	Bio-rad	MCA1076GA	Poor resolution
CD45RO	UCHL1	Pre-conjugated	¹⁶³ Dy	Surface	Human	Standard Biotools	3163032C	No staining detected
CD8a	SK1	Pre-conjugated	¹⁶⁸ Er	Surface	Human	Standard Biotools	3168002C	No staining detected
CD79a	HM57	In-house conjugated	¹⁶⁹ Tm	Intracellular	Human	Bio-rad	MCA2538GA	Poor specificity
CD63	H5C6	In-house conjugated	¹⁷⁰ Er	Surface	Human	Biotechnie	NBP2-42225	No staining detected
HLA-DR	L243	Pre-conjugated	¹⁷³ Yb	Surface	Human	Standard Biotools	3173005C	Weak staining

TABLE 7 | Origin of canine PBMCs donors.

Donor	Small Animal Department, Ghent University	Commercially available source (iQBiosciences)
D1	x	
D2	x	
D3	x	
D4	x	
D5970		x

with CD3 in a bivariate plot to distinguish B and T cells during initial gating (Figure 6).

We evaluated an alternative antibody for canine MHC-II detection: a commercially available anti-human HLA-DR

(Clone L243, ¹⁷³Yb), which has been reported to exhibit cross-reactivity with canine MHC class II [33]. Despite testing at elevated concentration, the signal intensity generated by anti-human HLA-DR Ab was consistently lower than that achieved with anti-canine MHC-II (Clone YKIX334.2, ¹⁷³Yb) (Figure 7A). Furthermore, the anti-canine MHC-II reliably discriminated classical monocytes and DCs as MHC-II positive populations; whereas this distinction was less clear with anti-human HLA-DR. Consequently, we have chosen anti-canine MHC-II to identify antigen-presenting cells in this panel due to its superior specificity and resolution.

For the identification of DCs, in addition to the well-characterized CD8a and MHC-II [23, 34], several other markers have been reported to be specific for DCs during maturation, including anti-human CD1a (Clone NA1/34, ¹⁵⁸Gd), anti-human CD40 (Clone LOB7/6, ¹⁶¹Dy), and anti-human CD86 (Clone Bu63, ¹⁵⁶Gd) [35–37]. In this study, our objective was to detect the major known leukocyte subsets present in normal canine

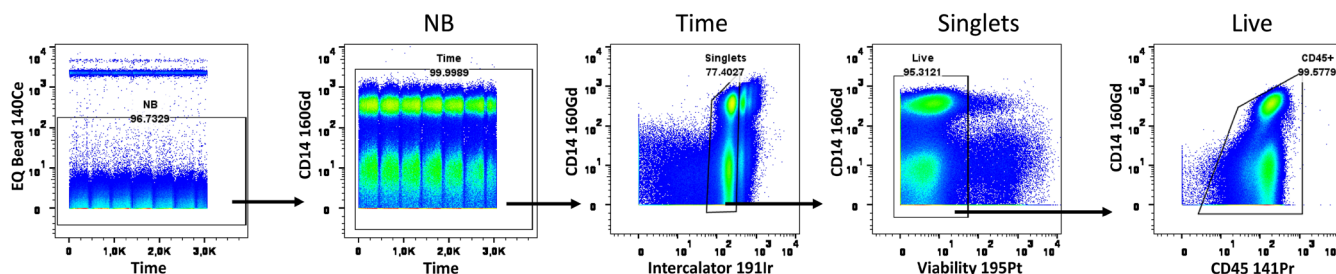


FIGURE 9 | Clean up gating strategy used for identifying viable CD45+ single cell population using monocyte marker CD14, ¹⁹¹Ir, and ¹⁹⁵Pt. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

PBMCs as comprehensively as possible. Therefore, we did not stimulate the PBMCs before analysis. As anticipated, expression of these maturation-associated markers was not observed in the uncultured DCs population (Figure 7B), consistent with previous findings by Bund et al. [38]. Additionally, CD11c (Clone CA11.6A1) has been described as a marker of immature myeloid DCs [39]. Since the purified antibody of this clone is not commercially available, we evaluated another clone of anti-human CD11c (Clone Bu15, ¹⁴⁷Sm), which has been reported to cross-react with canine cells [40], for the identification of immature myeloid DC population. However, no specific staining of DCs was observed with this antibody (Figure 7B). Taken together, these limitations necessitated the identification of DCs based on CD8 and MHC-II expression, following the exclusion of NKT cell, CD8+ T cell, and the CD8+ NK cell subset through a defined gating strategy.

For the detection of basophils, we found no published studies describing specific markers for canine basophils. In humans, CD63 and CD123 are established as reliable phenotyping markers for identifying basophils in their activated and resting states, respectively [41]. According to vendor information, anti-human CD63 (Clone H5C6, ¹⁷⁰Er) and anti-human CD123 (Clone 107D2.08, ¹⁵⁴Sm) are reported to cross-react with canine antigen. Therefore, we evaluated whether these Abs could be used to identify the canine basophil population. After exclusion of all known leukocyte populations, no specific staining was observed with anti-human CD63 nor anti-human CD123.

In dogs, the optimal marker scheme for defining memory T cell populations remains ambiguous. Previous studies have employed both CD44/CD62L and CD45RA/CD62L phenotypic paradigms to define memory subsets [4, 10, 29, 42]. However, as the canine-specific purified anti-CD45RA antibody (Clone CA4.1D3) is not commercially available, this marker could not be incorporated into our final panel. Furthermore, the anti-human CD62L (Clone FMC46, ¹⁴⁵Nd or ¹⁶²Dy) failed to discriminate between naïve/central memory and effector memory T cell subsets (Figure 8). We also evaluated CD45RO (Clone UCHL1, ¹⁶³Dy), a marker reportedly cross-reactive with canine central/effector memory T cells [33, 43], but it failed to yield specific staining in our experiments. Therefore, we substituted these markers with Eomesodermin, a T-box transcription factor best known for its role in cytotoxic lymphocyte differentiation, but which also exerts important and context-dependent functions in CD4+ T cells [8, 9, 11]. Eomes is transcriptionally upregulated during T cell activation and

effector differentiation, with highest expression in effector memory CD8+ T cells and long-lived memory precursors, distinguishing them from naïve or central memory cells that express lower levels [9, 13]. In parallel, CD44 has been functionally validated as an antigen-experience marker in canine peripheral blood where CD44-defined T cell subsets display distinct cytokine profiles; CD44 is upregulated following antigenic/mitogenic activation and remains elevated on memory CD4+ and CD8+ T cells, which produce higher levels of effector cytokines such as TNF- α compared with CD44-low naïve cells [4, 14]. As Eomes expression is highest in effector memory T cells, we used its co-expression with CD44, a well-established marker of antigen-experienced cells whose expression shifts from naïve (CD44-) to memory (CD44+) phenotype upon TCR engagement, to resolve functionally distinct subsets [10, 14]. For consistent identification of T cell subsets, we used bivariate plot CD44 versus Eomes to delineate $T_{\text{naïve}}$ (CD44-Eomes-), $T_{\text{central memory}}$ (CD44+Eomes-), and $T_{\text{effector-memory}}$ (CD44+Eomes+) populations. Using this approach, we observed that CD8+ T cells have greater effector memory populations compared to CD4+ T cells, reflecting their potent cytotoxic role (Figure 8).

The phenotypic definition of leukocyte subpopulations is listed in Table 5. Table 6 lists all antibodies tested but not used for this OMIP. By the use of mass cytometry, several channels remain unoccupied (e.g., ⁸⁹Y, ¹⁰², ¹⁰⁴, ¹⁰⁵Pd, ¹⁰⁶, ¹¹⁰–¹¹⁴, ¹¹⁶Cd, ¹⁴²–¹⁴⁶, ¹⁴⁸Nd, ¹⁴⁷, ¹⁴⁹, ¹⁵⁴Sm, ¹⁵⁵–¹⁵⁸Gd, ¹⁶¹–¹⁶²Dy, ¹⁶⁶, ¹⁶⁸, ¹⁷⁰Er, ¹⁶⁹Tm, ¹⁷², ¹⁷⁴Yb), providing flexibility for future panel expansion without extensive panel design or modification of the backbone itself. However, when assigning additional markers, it is important to avoid placing low-abundance targets in the $M \pm 1$ or $M + 16$ channels of isotopes assigned to highly expressed markers, as signal spillover in these positions may compromise accurate detection.

3.2.3 | Blood Sample

Blood samples were obtained from healthy purpose-bred dogs of the Small Animal Department, Ghent University. The ethics committee of the Ghent University approved the study (number EC2024-081). In addition, we also tested the commercially available Beagle Canine PBMCs source (IQB-CPB102liQBiosciences). The origin of all dog donors is detailed in Table 7. Depending on the availability of canine PBMC sources, specific dog donors were selected for individual experiments, as indicated in the respective figure legends.

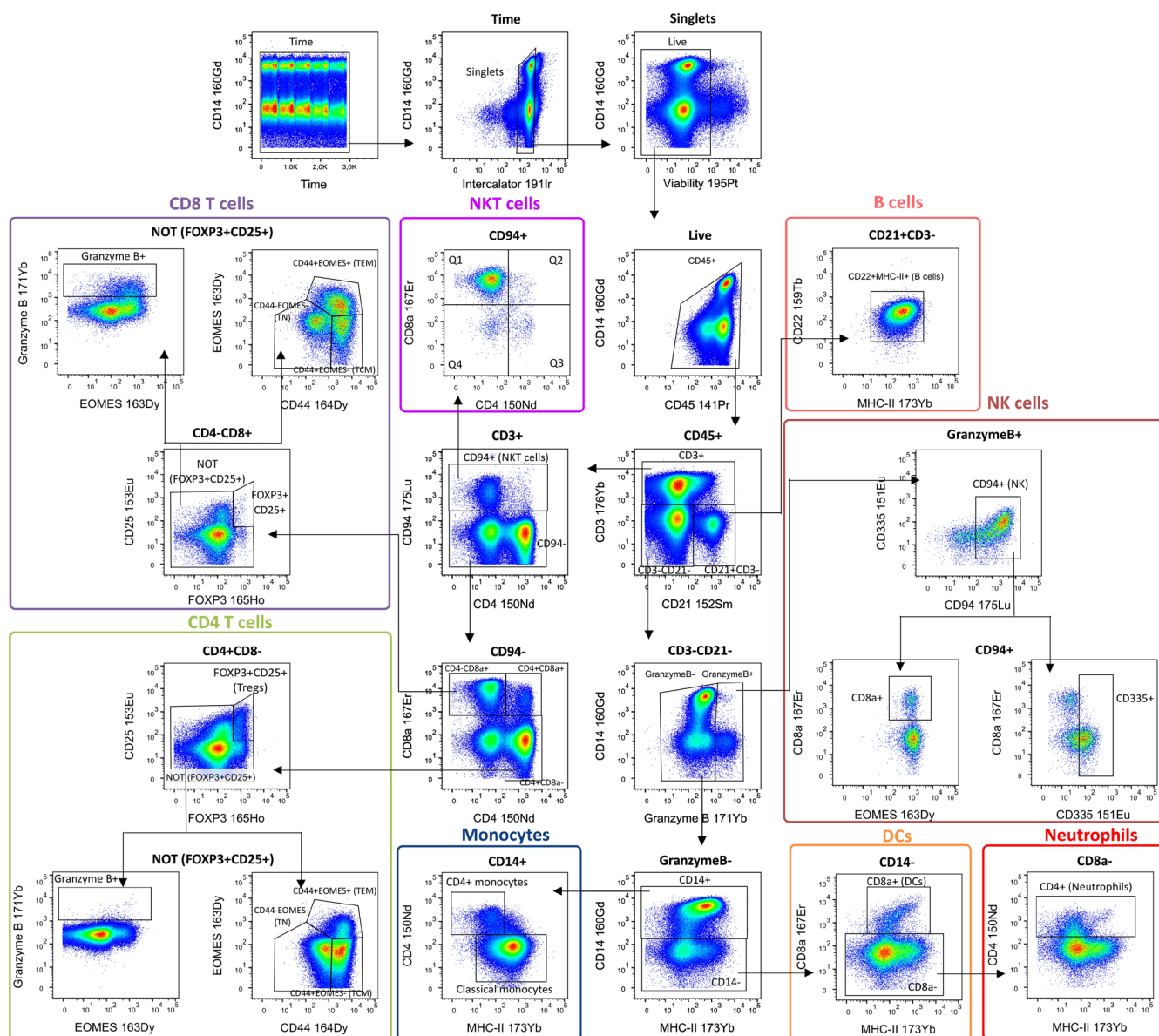


FIGURE 10 | Hierarchical gating strategy. Representative example of manual gating strategy for cryopreserved canine PBMCs using Lunatron mass cytometer. First, cell events were gated on any marker versus time to ensure data quality and consistency. Singlets were then gated using a bivariate plot of the DNA intercalator versus monocyte marker CD14. Subsequently, live cells were identified by gating out ^{195}Pt positive cells. CD45 ^{141}Pr were then used to identify leukocyte identity. T and B cells were subsequently identified using CD3 and CD21, respectively. B cell identity was further confirmed by CD22 and MHC-II expression (CD22+MHC-II+). Within the CD3+ population, NKT cells were defined as CD94+, whereas conventional T cells were defined as CD94-. Expression of CD4 and CD8 was assessed on NKT cells. Conventional T cells were further classified into single positive CD4+ T cells (CD4+CD8-), single positive CD8+ T cells (CD4-CD8+), and double positive (CD4+CD8+) subsets. In both CD4+CD8- and CD4-CD8+ T cells, regulatory T cells were identified as CD25+FOXP3+. Within the NOT (FOXP3+CD25+) population, CD44 and EOMES were used to classify naïve (CD44-EOMES-), central-memory (CD44+EOMES-), and effector-memory (CD44+EOMES+) subsets among CD4 and CD8 T cells. Expression of cytotoxic activity (Granzyme B) was also assessed on CD4 and CD8 conventional T cells. B cells were identified as CD21+CD22+MHC-II+. Within the CD3-CD21- population, NK cells were defined as Granzyme B+ and further characterized based on the expression of CD94. Expression of CD8a and CD335 was also assessed on NK cells. Within the Granzyme B- population, monocytes were identified as CD14+ and further subdivided into CD4+ monocytes (CD4+MHC-II-) and classical monocytes (CD4-MHC-II+). Within the CD14- population, DCs were defined as CD8+MHC-II+. Within the CD8- population, neutrophils were identified as CD4+. Hierarchical gating was performed using D3. Mass cytometry data were exported as FCS file using ArcSinh transformation with a cofactor of 5. Analysis of mass cytometry data was carried out using FlowJo v10.10. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

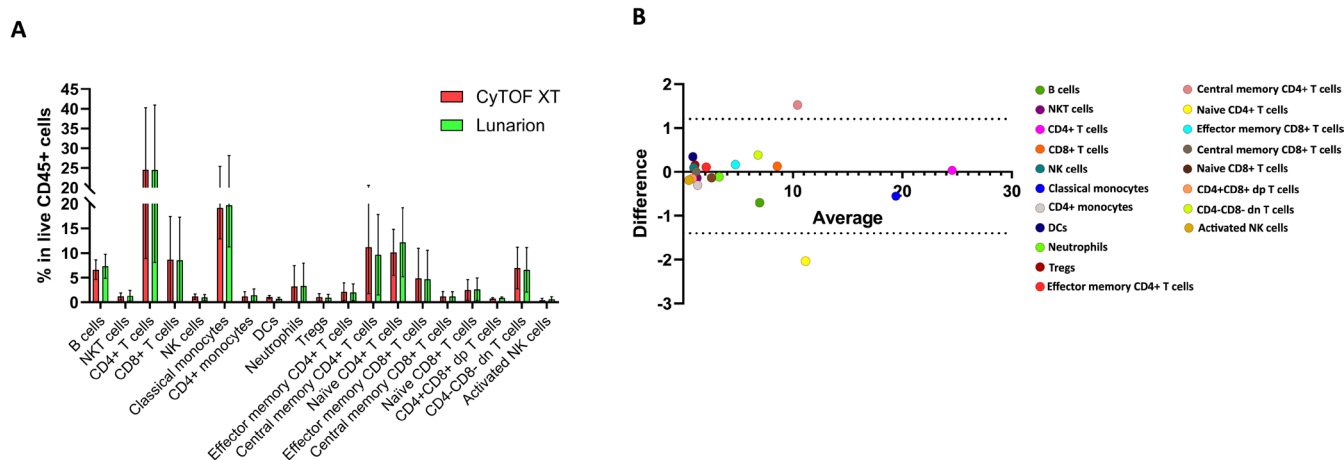


FIGURE 11 | Comparison of the CyTOF XT and Lunarion instruments. (A) Comparison of the frequencies of reported leukocyte subsets within live single CD45+ cells, as assessed by both the CyTOF XT and Lunarion instruments. The results are shown as the mean $\pm$ SD ($n=3$ including D1, D3, D5970). (B) Bland–Altman analysis of reported leukocyte subset frequencies within live single CD45+ cells, as assessed by both the CyTOF XT and Lunarion instruments ($n=3$ per subset including D1, D3, D5970). Mean bias was -0.0714% (SD = 0.6553%), with 95% limits of agreement from -1.356% to $+1.213\%$. Dashed lines indicate limits of agreement. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

3.3 | Data Preprocessing, Controls and Other Technical Considerations

3.3.1 | Data Clean up, Normalization and Analysis

Following data acquisition, EQ Six Element Calibration Beads were utilized for normalization, and standardized FCS files were generated for downstream analysis. First, cells were identified by gating out the EQ Six Element Calibration Beads. Next, cell events were gated on any marker versus time to ensure data quality and consistency. Singlets were then gated using a bivariate plot of the DNA intercalator versus monocyte marker CD14, as monocytes exhibit elevated nucleic acid content, resulting in increased intercalator uptake. Gating based on the ^{191}Ir and ^{193}Ir axis may partially exclude monocytes [44]. Subsequently, live cells were identified by gating out ^{195}Pt positive cells. CD45 ^{141}Pr were then used to identify leukocyte identity (Figure 9). Finally, we manually gated our desired lineage population (Table 5). The overall lineage gating strategy on CyTOFXT is illustrated in Figure 1. The full gating strategy on the Lunarion instrument is shown in Figure 10. Mass cytometry data were exported as FCS files using ArcSinh transformation with a cofactor of 4 for CyTOF XT and a cofactor of 5 for Lunarion. Analysis of mass cytometry data was carried out using FlowJo v10.10. Statistical analyses were conducted using GraphPad Prism 10.

A comparison of reported leukocyte subsets frequencies within live single CD45+ cells, as assessed by both the CyTOF XT and Lunarion instruments, is illustrated in Figure 11A. Agreement between CyTOF XT and Lunarion was assessed by Bland–Altman analysis (Figure 11B). In general, Bland–Altman analysis demonstrated excellent agreement between CyTOF XT and Lunarion across leukocyte subsets (mean bias = -0.0714% , 95% limits of agreement = $[-1.356\%, 1.213\%]$), with the exception of central memory and naïve CD4+ T cell subsets. Greater variability in these two subset frequencies reflects their adjacent position in the bivariate plot of secondary markers (Eomes and CD44), where small threshold

adjustments cause reciprocal shifts in measured frequencies [45]. However, this cross-platform comparison is based on a limited sample size ($n=3$), which constrains the strength of the conclusions. With only three biological replicates, the estimated bias and limits of agreement should be interpreted as preliminary, and may not fully capture inter-individual variability or rare sources of technical discordance between instruments. Consequently, the current data support the notion of good agreement between CyTOF XT and Lunarion for the leukocyte subsets examined, but more extensive validation in a larger cohort would be required to robustly establish cross-platform equivalence.

3.3.2 | Other Technical Considerations

We noted that the intracellular markers (FoxP3, Granzyme B, and Eomes) exhibited relatively high background staining, which constrained the resolution of clearly defined positive populations. This nonspecific signal may arise from factors such as fixation/permeabilization artifacts or antibody cross-reactivity with fixation-altered epitopes. The resolution of positive populations also varies depending on the instrument. Although these markers remained interpretable within the context of our gating strategy, further optimization such as evaluation of alternative fixation/permeabilization buffers or testing additional antibody clones could improve signal-to-noise ratios and strengthen the reproducibility of these readouts in future applications.

In summary, we developed a 15-marker mass cytometry panel for the comprehensive characterization of the major leukocyte subsets within canine PBMCs. Importantly, the panel retains flexibility for future expansion through several unassigned metal channels, paving the way for future expansion. This OMIP provides a robust and scalable framework to support high-dimensional immune monitoring in canine models, thereby advancing translational research [1], improving the predictive power of preclinical safety and efficacy studies

[46], and enabling detailed immune profiling in various disease states [47].

Author Contributions

Dominique Revets: methodology, writing – review and editing. **Antonio Cosma:** conceptualization, methodology, data curation, formal analysis, writing – review and editing, funding acquisition, resources, investigation. **Ario de Marco:** writing – review and editing, funding acquisition. **Huyen Thuc Tran Luong:** methodology, data curation, investigation, formal analysis, visualization, writing – original draft, writing – review and editing, resources. **Sofie Vercammen:** writing – review and editing, resources. **Hilde de Rooster:** writing – review and editing, resources, funding acquisition.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in FlowRepository at <http://flowrepository.org/>, reference number FR-FCM-Z9FA.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** cytoa70040-sup-0001-Supinfo.doc.